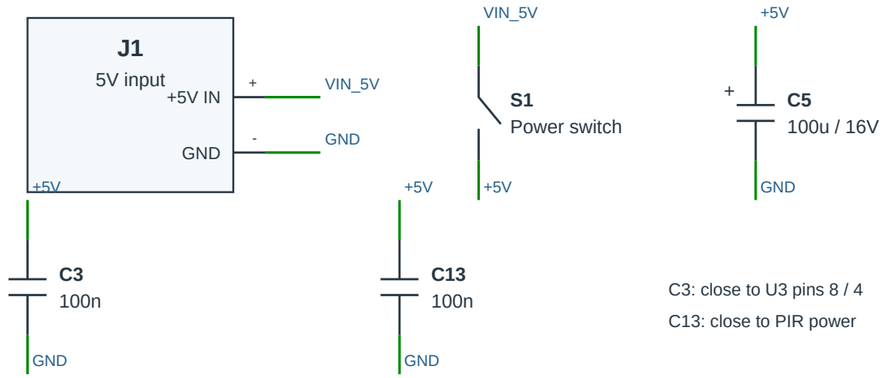
LED lighthouse / 01 Light sensor and power

LDR5516 photoresistor, TLC372CP dual comparator, daylight inhibit and common 5V power.

A / Power input and local bypass



B / Supply notes and behaviour

Use regulated +5V from USB or a verified boost output.

Do not connect a 3.7V cell to VIN_5V. All GND nodes join.

Battery, charger and boost are external to this schematic.

PIR connector: match VCC / OUT / GND labels on your board.

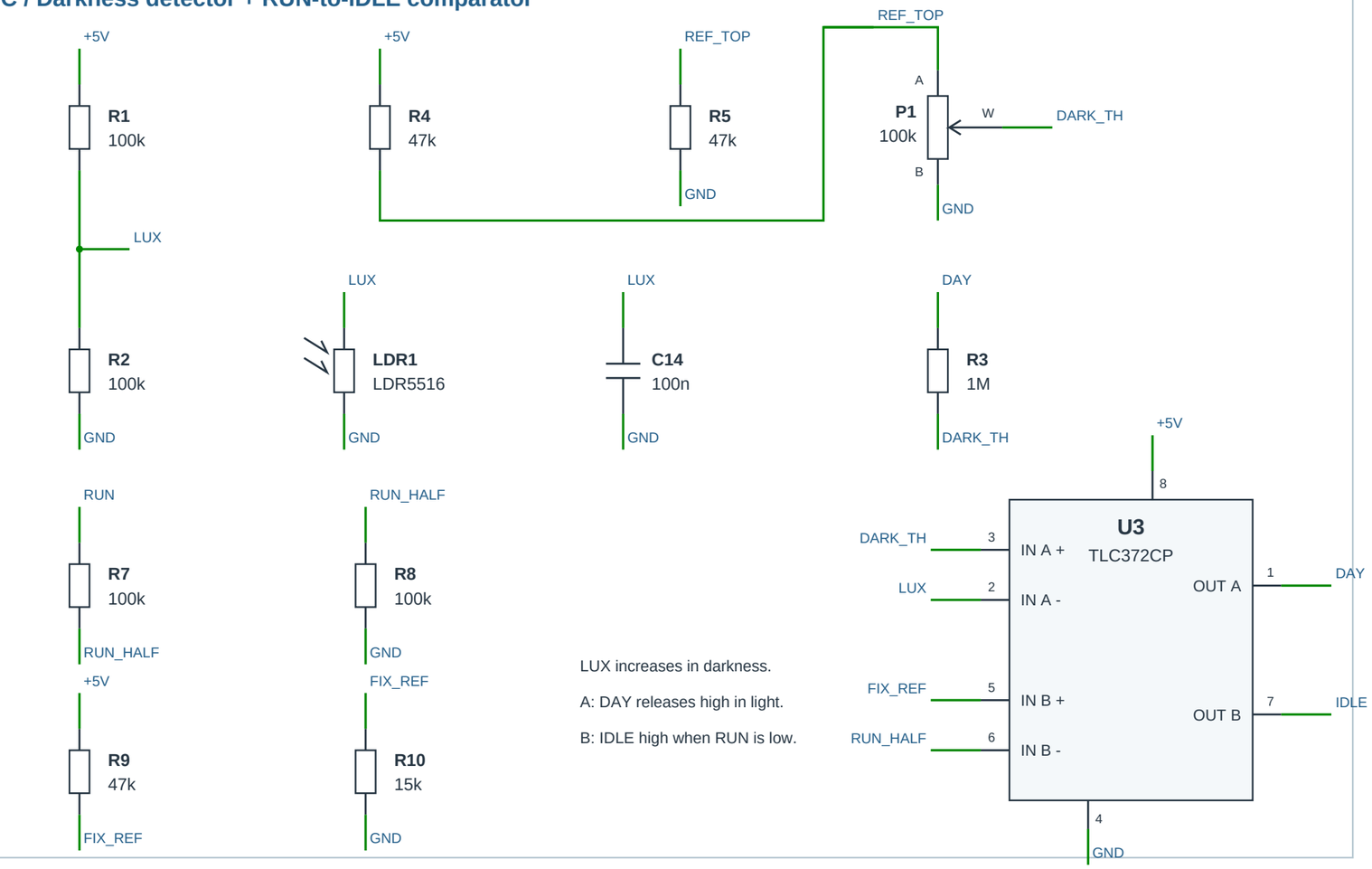
Dark + a NEW motion edge starts one run; daylight cancels it.

Shield the LDR from LED light. Adjust P1 at the desired dusk level.

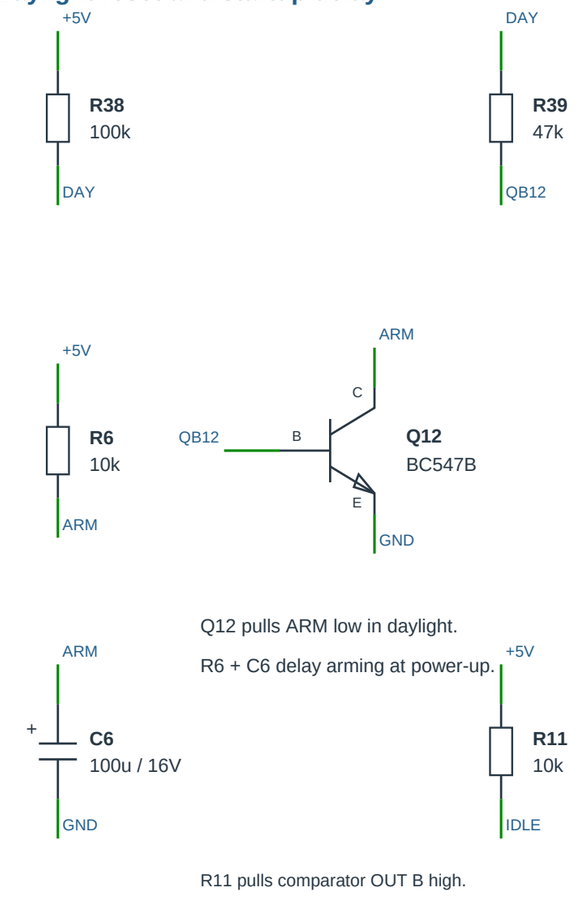
Only the manual five-LED CD4017 test is user-confirmed.

New sensor / timer stages need the staged bench checks.

C / Darkness detector + RUN-to-IDLE comparator



D / Daylight reset and startup delay



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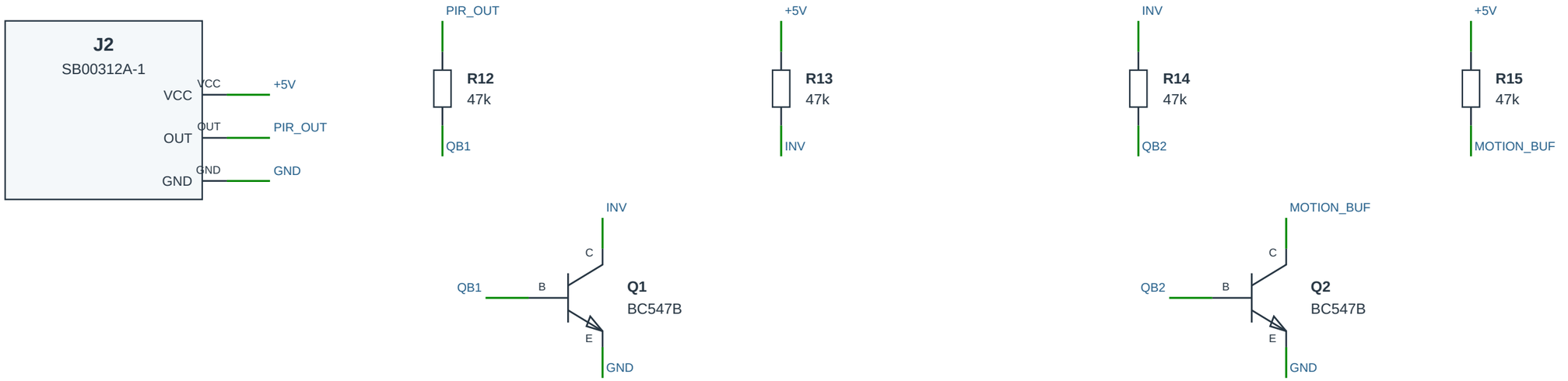
Same net name = direct electrical connection, including between sheets. Pin positions are functional, not a DIP top view.

LED lighthouse / 02 Motion pulse and timeout

A buffered PIR edge triggers a TLC555 one-shot. Busy lockout ignores new edges during RUN.

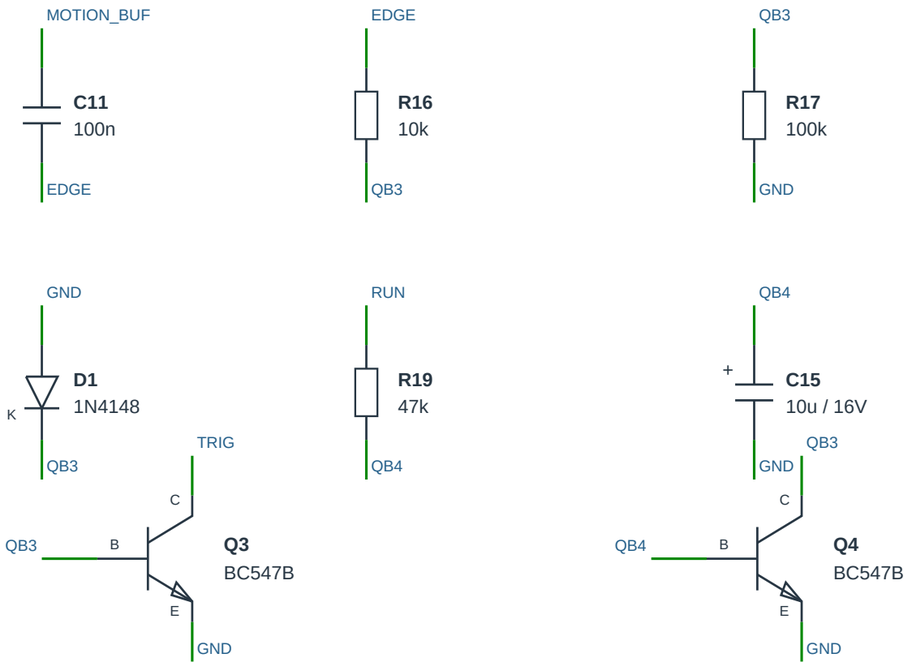
E / PIR input and edge pulse (SB00312A-1 signal names; no physical pad order assumed)



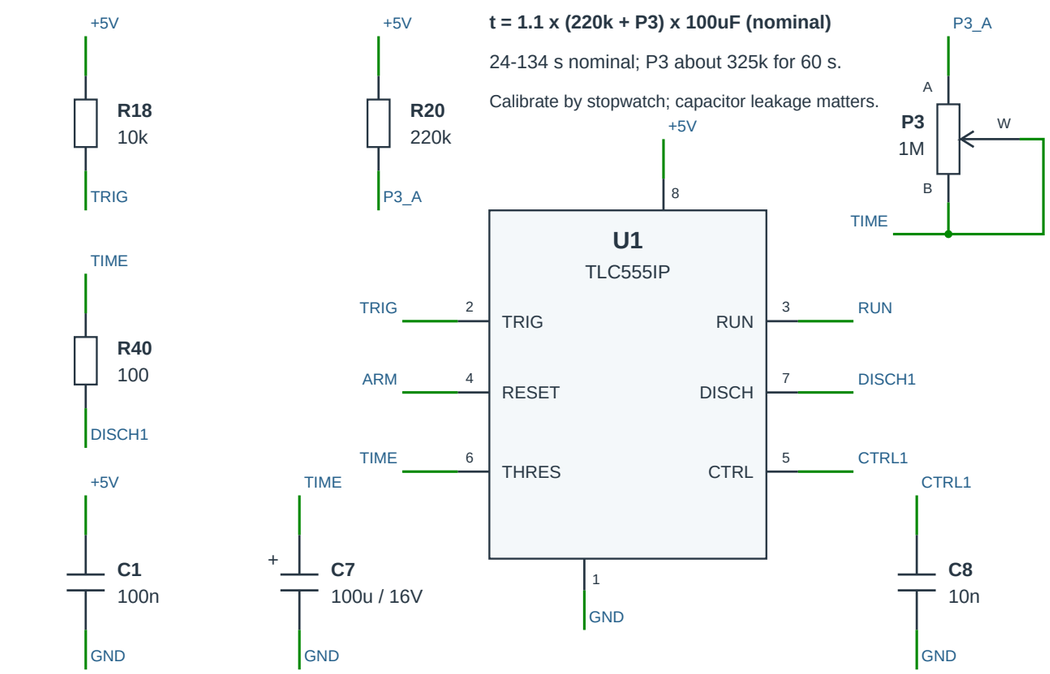
Q1 inverts PIR_OUT; Q2 restores a positive 5V edge. C11 couples only the change, not a held PIR HIGH.

Confirm LOW at rest / HIGH on motion on the actual PIR. Startup/warmup motion pulses may start one run.

F / Short trigger + busy lockout



G / Approximately one minute (TLC555IP)



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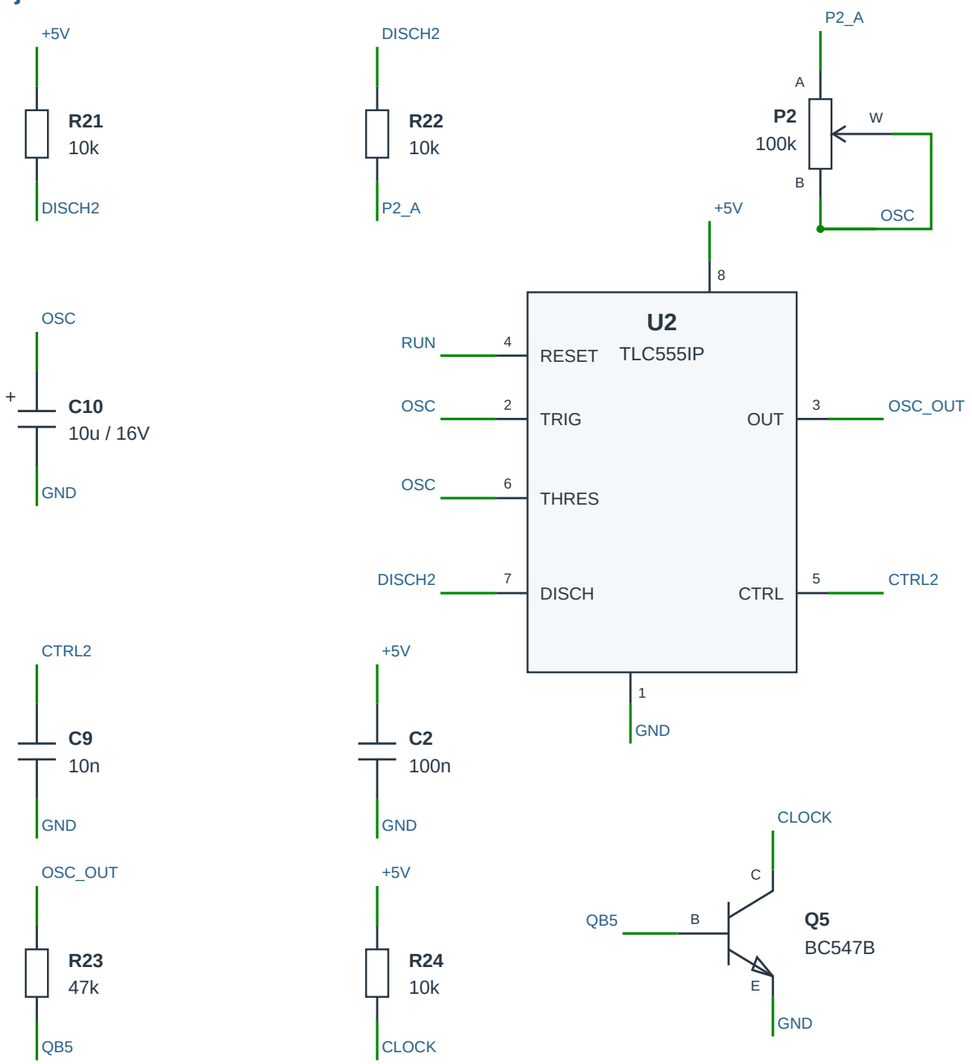
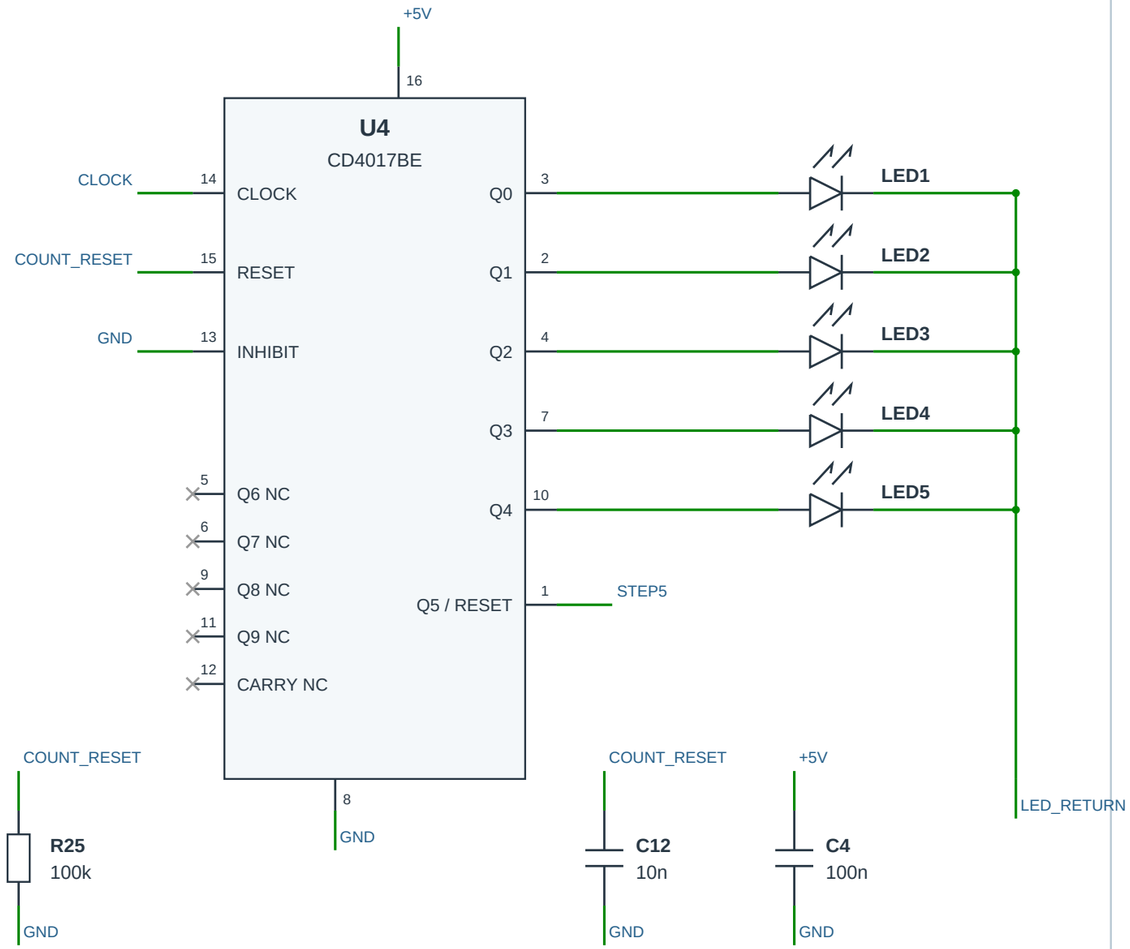
LED lighthouse / 03 Running LEDs and complete blanking

One TLC555 oscillator + CD4017 modulo-five sequencer. Q6 switches the common LED return.

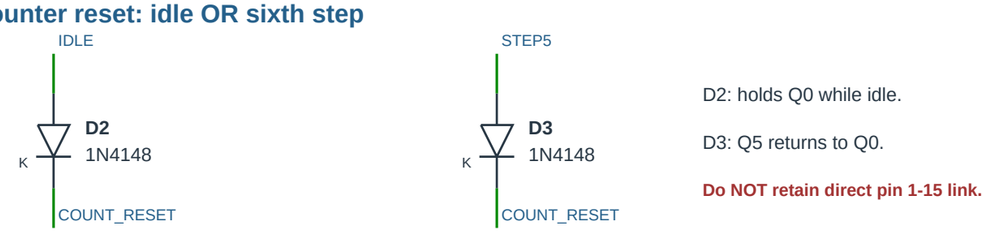
H / Adjustable oscillator and inverted clock



I / CD4017 and five direct-drive white LEDs



J / Counter reset: idle OR sixth step

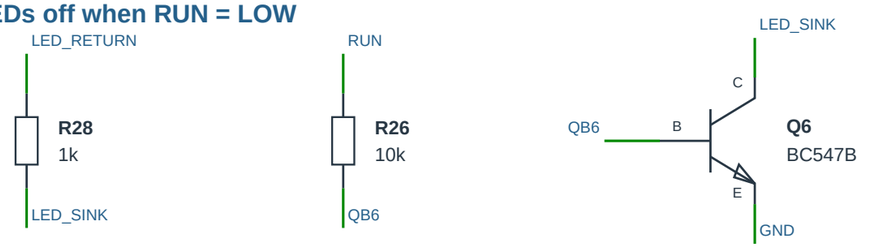


D2: holds Q0 while idle.

D3: Q5 returns to Q0.

Do NOT retain direct pin 1-15 link.

K / All LEDs off when RUN = LOW



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